

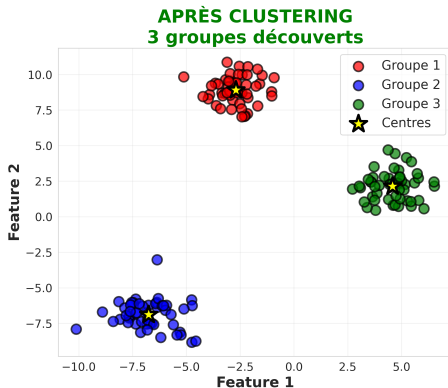
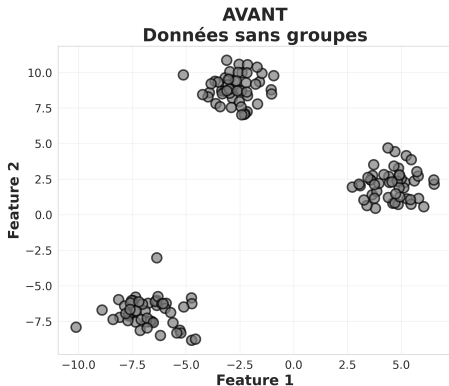
Clustering

A Short Introduction

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Academic Year: 2025 - 2026

What is Clustering?



Simple Definition

Clustering = Grouping

Dividing data into groups where the points resemble each other

Goal:

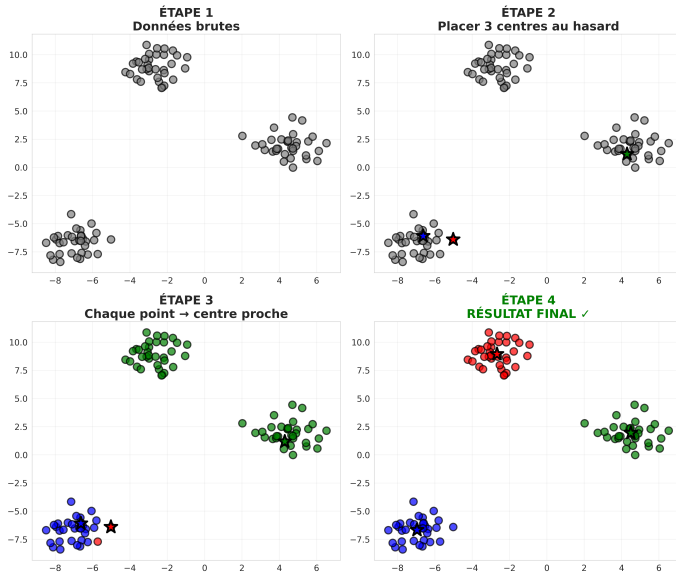
- Points in the same group → **similar**
- Points in different groups → **different**

No labels!

The algorithm discovers the hidden groups in the data on its own

K-Means: The Main Algorithm

ALGORITHME K-MEANS



How Does K-Means Work?

Principle

K-Means finds K groups by placing centers

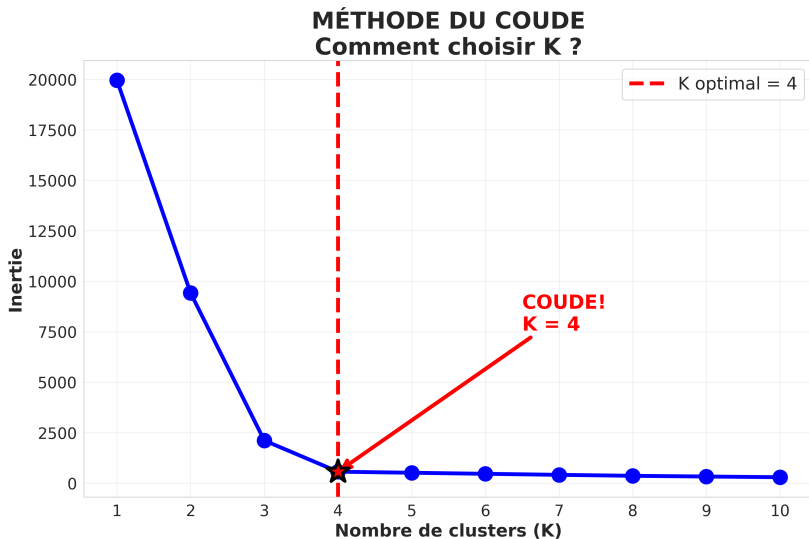
The 4 steps:

- 1 Choose K (desired number of groups)
- 2 Place K centers at random
- 3 Assign each point to the nearest center
- 4 Repeat until the centers stop moving

Example

To segment customers: $K=3 \rightarrow$ Budget, Mid-range, Premium groups

How Do We Choose K?



The Elbow Method

Problem

K-Means needs to know K in advance. How do we choose it?

Solution: Look for the "elbow"

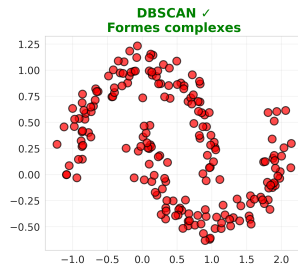
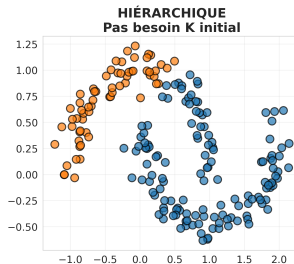
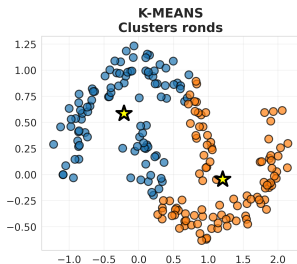
- Test several values of K (1, 2, 3, ..., 10)
- Measure the quality for each K
- The plot forms an "elbow"
- The elbow indicates the optimal K

Rule

Choose K at the elbow = good trade-off!

The 3 Clustering Algorithms

COMPARAISON DES 3 ALGORITHMES



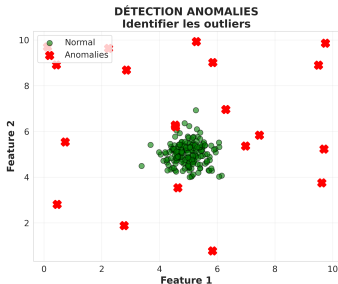
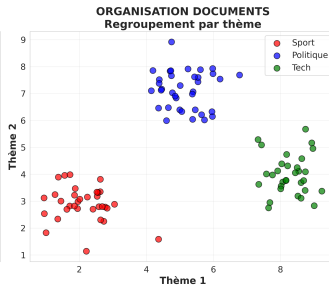
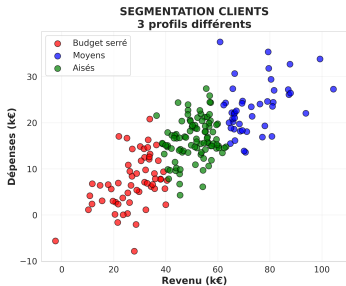
Comparing the 3 Algorithms

Algorithm	Advantage	Use
K-Means	Fast and simple	Round groups
Hierarchical	No need for K	Small datasets
DBSCAN	Complex shapes	Noise/Outliers

Which to choose?

- **K-Means:** Most of the time (fast, effective)
- **Hierarchical:** When you don't know K
- **DBSCAN:** Irregular shapes or noise present

APPLICATIONS DU CLUSTERING



ET BIEN PLUS...

AUTRES APPLICATIONS

- Compression images (réduire couleurs)
- Recommandations (clients similaires)
 - Biologie (groupes de gènes)
- Réseaux sociaux (communautés)

Examples of Use

Domain	Application
Marketing	Customer segmentation
E-commerce	Product recommendations
Documents	Organization by topic
Security	Anomaly detection
Images	Compression (reducing colors)
Biology	Gene groups
Social networks	Communities

Typical Case

A company with 10,000 customers → 4 segments for targeted marketing

How to Do Clustering? (5 Steps)

- 1 **Collect** the data
- 2 **Normalize** (bring to the same scale)
- 3 **Choose K** (elbow method)
- 4 **Apply K-Means**
- 5 **Interpret** the groups

The Most Important Thing

Interpretation! The groups must make sense for your domain.

Key Points to Remember

1. UNsupervised Learning

No labels → we discover hidden structures

2. K-Means = The Main Algorithm

Fast, simple, effective for most cases

3. Elbow Method

To find the right number of groups (K)

4. Normalize the Data

ALWAYS bring the features to the same scale

5. Applications Everywhere

Marketing, documents, anomalies, compression, etc.

Clustering is:

- Finding groups in unlabeled data
- K-Means: the main tool (fast and effective)
- Elbow method: for choosing K
- Many applications: customers, documents, etc.

The Essential Takeaway

Clustering reveals hidden structures to help you better understand your data